

PLSC 40601

Week 7: Efficient estimation.

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Spring 2026

Some asymptotic language.

Convergence

Review [Lehmann \(1999\)](#) Chapter 2

- A deterministic sequence a_n converges to a , written

$$a_n \rightarrow a,$$

if:

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - a| < \varepsilon$.

- I.e., for every tolerance ε , no matter how small, there is some point N after which every term of the sequence is within ε of the limit a .

Convergence in probability

- A sequence of random variables X_n converges in probability to X if, for every $\varepsilon > 0$,

$$\Pr[|X_n - X| > \varepsilon] \rightarrow 0.$$

- We write

$$X_n \xrightarrow{p} X.$$

- For an estimator:

$$\hat{\theta}_n \xrightarrow{p} \theta_0$$

means $\hat{\theta}_n$ is consistent for θ_0 .

Convergence in distribution

- A sequence X_n converges in distribution to X if

$$F_n(x) \rightarrow F(x)$$

at every continuity point x of the limiting distribution function F .

- We write

$$X_n \xrightarrow{d} X.$$

OR

$$X_n \rightsquigarrow X.$$

- E.g., for asymptotic normality:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, V).$$

- Intuition: the shape of the sampling distribution stabilizes.

Big-oh and little-oh

- For deterministic sequences, $a_n = O(b_n)$ means

$$\left| \frac{a_n}{b_n} \right|$$

is eventually bounded.

- $a_n = o(b_n)$ means

$$\frac{a_n}{b_n} \rightarrow 0.$$

- So:

$$a_n = O(n^{-1/2})$$

means a_n is no larger than root- n order.

- And:

$$a_n = o(n^{-1/2})$$

means a_n is negligible relative to root- n order.

Stochastic order notation

- For random sequences, $X_n = O_p(a_n)$ means

$$\frac{X_n}{a_n}$$

is bounded in probability.

- $X_n = o_p(a_n)$ means

$$\frac{X_n}{a_n} \xrightarrow{p} 0.$$

- Examples:

$$\hat{\theta}_n - \theta_0 = O_p(n^{-1/2})$$

means root- n consistency.

-

$$\hat{\theta}_n - \theta_0 = o_p(1)$$

means ordinary consistency.

Why rates matter here

- Efficient-estimation arguments separate a leading term from a remainder:

$$\hat{\psi} - \psi_0 = \text{first-order term} + \text{remainder}.$$

- We want the remainder to be negligible:

$$\text{remainder} = o_p(n^{-1/2}).$$

- In doubly robust estimators, the remainder often behaves like a product of nuisance errors:

$$\|\hat{m} - m_0\| \|\hat{e} - e_0\|.$$

- That product can be small enough even when each nuisance estimate is slower than root- n .

The empirical measure

- The empirical measure puts mass $1/n$ on each observed data point:

$$\mathbb{P}_n = \frac{1}{n} \sum_{t=1}^n \delta_{O_t}.$$

- For a set A :

$$\mathbb{P}_n[A] = \frac{1}{n} \sum_{t=1}^n \mathbb{1}(O_t \in A).$$

- For a function f :

$$\mathbb{P}_n f = \frac{1}{n} \sum_{t=1}^n f(O_t), \quad P_0 f = \mathbb{E}_{P_0}[f(O)].$$

- This lets us write sample averages as integrals with respect to $\mathbb{P}_n$.

Empirical-process notation

- The gap between sample and population averages is

$$(\mathbb{P}_n - P_0)f = \frac{1}{n} \sum_{t=1}^n f(O_t) - \mathbb{E}_{P_0}[f(O)].$$

- The central limit theorem gives, for fixed f ,

$$\sqrt{n}(\mathbb{P}_n - P_0)f \xrightarrow{d} N\{0, \text{Var}_{P_0}[f(O)]\}.$$

- Influence-function estimators are built so their leading term has this form.

Model classes.

What is a statistical model?

- We observe

$$Z_1, \dots, Z_n \stackrel{iid}{\sim} P_0,$$

where P_0 is the unknown true data-generating distribution.

- A **statistical model** is a set of distributions:

$$\mathcal{P} = \{P : P \text{ is allowed by our assumptions}\}.$$

- Saying

$$P_0 \in \mathcal{P}$$

means the true distribution is one of the distributions allowed by the model.

A model is a set of possible worlds

- Here, $\mathcal{a}$ is a mathematical set of possible worlds.
- The stronger the assumptions, the smaller the set $\mathcal{P}$.
- The weaker the assumptions, the larger the set $\mathcal{P}$.

Three kinds of models

- **Parametric:** $\mathcal{P}$ is indexed by finitely many parameters.
- **Nonparametric:** $\mathcal{P}$ is infinite-dimensional, often very large.
- **Semiparametric:** the target is finite-dimensional, but the nuisance structure is infinite-dimensional.

Parametric models

- A **parametric model** is indexed by a finite-dimensional parameter:

$$\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}^d\}.$$

- Once θ is known, the whole distribution is known.
- Normal example:

$$\mathcal{P} = \{N(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}.$$

- Saying $P_0 \in \mathcal{P}$ means the true distribution is normal for some μ and σ^2 .
- Parametric models are restrictive, but they give sharp benchmarks like Fisher information.

Semiparametric models

- A **semiparametric model** has a finite-dimensional target and infinite-dimensional nuisance components.

- Example target:

$$\psi_0 = \mathbb{E}[Y(1) - Y(0)].$$

- Under identification, $\psi_0 = \Psi(P_0)$, but P_0 contains nuisance functions such as

$$\mu_a(x) = \mathbb{E}[Y \mid A = a, X = x], \quad e(x) = \Pr[A = 1 \mid X = x].$$

- We care about ψ_0 , but we may estimate μ_a and e flexibly with ML.

Regression restrictions: semiparametric

- Some models restrict part of the distribution rather than every feature of it.
- Example:

$$\mathbb{E}_{P_0}[Y \mid X] = \beta_0 + \beta_1 X.$$

- Then $\mathcal{P}$ is the set of distributions whose conditional mean of Y given X is linear.
- Saying $P_0 \in \mathcal{P}$ means the true conditional mean is linear.
- This is usually **semiparametric**: β is finite-dimensional, but the distribution of X , the error distribution, and higher moments remain unrestricted.

When regression becomes parametric

- To make the regression model fully parametric, we would need stronger assumptions, e.g.

$$Y \mid X \sim N(\beta_0 + \beta_1 X, \sigma^2).$$

- Then a finite-dimensional parameter determines the full conditional distribution:

$$\theta = (\beta_0, \beta_1, \sigma^2).$$

Nonparametric models

- A **nonparametric model** allows the distribution to vary over a much larger, infinite-dimensional class.
- In the simplest case:

$$\mathcal{P} = \{P : P \text{ is any probability distribution on the sample space}\}.$$

- Then $P_0 \in \mathcal{P}$ is very weak: it only says the truth is a valid probability distribution.

Nonparametric restrictions

- Example:

$$\mathcal{P} = \{P : \mathbb{E}_P[Y^2] < \infty\}.$$

- The model does not say Y is normal, linear in X , logistic, or indexed by a finite number of coefficients.
- This flexibility is useful, but it usually makes estimation and inference harder.
- Later restrictions may add smoothness or sparsity assumptions.

Functionals

- A **functional** is a function whose input is itself a whole distribution, curve, density, regression function, or other infinite-dimensional object.

- Ordinary function:

$$f : x \mapsto f(x).$$

- Functional:

$$\Psi : P \mapsto \Psi(P).$$

- In semiparametric causal inference, the target is usually a functional of the observed-data distribution:

$$\psi_0 = \Psi(P_0).$$

Function versus functional

- $\mu(x)$ is a function:

$$x \mapsto \mu(x).$$

- It takes a covariate value x as input and returns a number.
- $\mathbb{E}[\mu(X)]$ is a functional of P :

$$\Psi(P) = \int \mu_P(x) dP_X(x).$$

- It depends on the whole regression function $\mu_P(\cdot)$ and on the distribution of X .

Why semiparametric language matters

- Causal ML usually does not ask us to estimate the whole distribution P_0 well.
- It asks us to estimate a lower-dimensional target $\Psi(P_0)$ well.
- The nuisance functions can be high-dimensional or nonparametric.
- Efficient estimation is about extracting first-order information for the target while reducing sensitivity to nuisance-estimation error.

Fisher information.

Why start with Fisher information?

- We want to estimate a target parameter θ from data $O_1, \dots, O_n$.
- Many estimators are consistent: they converge to the right answer.
- Efficiency asks a sharper question: among regular estimators, how small can the asymptotic variance be?
- Fisher information gives the cleanest version of this question in parametric models. Lehmann (1999) Section 7.2

Parametric model

- Suppose observations are i.i.d.:

$$O_1, \dots, O_n \sim P_{\theta_0}, \quad \theta_0 \in \Theta \subset \mathbb{R}.$$

- The model says the whole distribution is indexed by one unknown number:

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

- If $p_\theta(o)$ is the density or mass function, the likelihood is

$$L_n(\theta) = \prod_{t=1}^n p_\theta(O_t),$$

and the log-likelihood is

$$\ell_n(\theta) = \sum_{t=1}^n \log p_\theta(O_t).$$

The score

- The score is the derivative of the log likelihood contribution:

$$s_{\theta}(O_t) = \frac{\partial}{\partial \theta} \log p_{\theta}(O_t).$$

- The sample score is

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta) = \sum_{t=1}^n s_{\theta}(O_t).$$

- The score tells us the local direction in which the data push the parameter estimate.

A key property

- Under regularity conditions, the score has mean zero at the true parameter:

$$\mathbb{E}_{\theta_0}[s_{\theta_0}(O)] = 0.$$

- Intuition:

$$\int p_{\theta}(o) do = 1 \quad \Rightarrow \quad \frac{\partial}{\partial \theta} \int p_{\theta}(o) do = 0.$$

- Since by the chain rule

$$\frac{\partial}{\partial \theta} p_{\theta}(o) = p_{\theta}(o) \frac{\partial}{\partial \theta} \log p_{\theta}(o),$$

the average score must be zero.

Fisher information

- Fisher information is the variance of the score.
- Since the score is mean zero, this variance can be written as

$$I(\theta) = \mathbb{E}_{\theta}[s_{\theta}(O)^2].$$

- Equivalently, under regularity conditions,

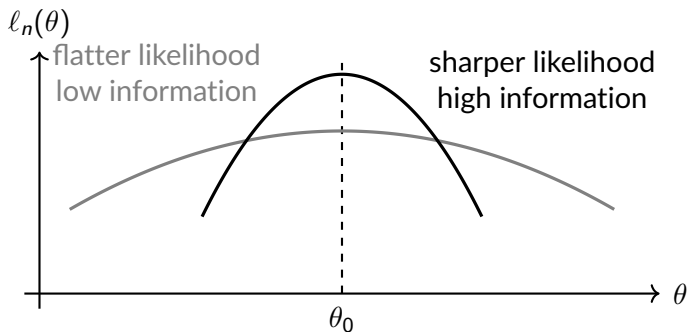
$$I(\theta) = -\mathbb{E}_{\theta} \left[\frac{\partial^2}{\partial \theta^2} \log p_{\theta}(O) \right].$$

- For n i.i.d. observations:

$$I_n(\theta) = nI(\theta).$$

- More information means the likelihood is more sharply curved around the truth.

Likelihood curvature



- Nearby values of θ are harder to distinguish when the likelihood is flat.
- Nearby values of θ are easier to distinguish when the likelihood is sharply curved.

Example: Bernoulli data

- Let $O_t \in \{0, 1\}$ and

$$P_\theta[O_t = 1] = \theta.$$

- Then the probability mass function is

$$p_\theta(O_t) = \theta^{O_t}(1 - \theta)^{1-O_t}.$$

- I.e.,:

$$O_t = 1 \Rightarrow p_\theta(O_t) = \theta, \quad O_t = 0 \Rightarrow p_\theta(O_t) = 1 - \theta.$$

Example: Bernoulli score

- Recall the score is the derivative of the log likelihood contribution:

$$s_{\theta}(O_t) = \frac{\partial}{\partial \theta} \log p_{\theta}(O_t).$$

- For Bernoulli data, the log likelihood contribution is

$$\log p_{\theta}(O_t) = O_t \log(\theta) + (1 - O_t) \log(1 - \theta).$$

- Differentiate with respect to θ :

$$s_{\theta}(O_t) = \frac{O_t}{\theta} - \frac{1 - O_t}{1 - \theta}.$$

- Put over a common denominator:

$$s_{\theta}(O_t) = \frac{O_t - \theta}{\theta(1 - \theta)}.$$

Bernoulli information: plug in the score

- Recall Fisher information is the variance of the score:

$$I(\theta) = \mathbb{E}_\theta[s_\theta(O_t)^2].$$

- We just found:

$$s_\theta(O_t) = \frac{O_t - \theta}{\theta(1 - \theta)}.$$

- Plug that score into the definition:

$$I(\theta) = \mathbb{E}_\theta \left[\left\{ \frac{O_t - \theta}{\theta(1 - \theta)} \right\}^2 \right].$$

- The denominator is not random, so it can move outside the expectation:

$$I(\theta) = \frac{\mathbb{E}_\theta[(O_t - \theta)^2]}{\{\theta(1 - \theta)\}^2}.$$

Bernoulli information: simplify

- Since $O_t \sim \text{Bernoulli}(\theta)$,

$$\mathbb{E}_\theta[O_t] = \theta, \quad \text{Var}_\theta[O_t] = \theta(1 - \theta).$$

- The numerator is the variance:

$$\mathbb{E}_\theta[(O_t - \theta)^2] = \text{Var}_\theta[O_t].$$

- Therefore:

$$I(\theta) = \frac{\text{Var}_\theta[O_t]}{\{\theta(1 - \theta)\}^2} = \frac{1}{\theta(1 - \theta)}.$$

- So the Fisher information for one observation is

$$I(\theta) = \frac{1}{\theta(1 - \theta)}.$$

Information and variance

- The MLE is $\hat{\theta} = \bar{O}$.
- By the central limit theorem, when $0 < \theta_0 < 1$:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\{0, \theta_0(1 - \theta_0)\}.$$

- Since

$$I(\theta_0)^{-1} = \theta_0(1 - \theta_0),$$

the MLE reaches the Fisher-information variance benchmark.

- This is the basic meaning of parametric efficiency.

Cramér–Rao intuition

- In a regular parametric model, no regular unbiased estimator can have variance below

$$\frac{1}{nI(\theta_0)}.$$

- More generally, regular asymptotically linear estimators face an asymptotic variance bound.
- So an efficient estimator is not just consistent.
- It is as stable as the model allows, to first order:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\{0, I(\theta_0)^{-1}\}.$$

Kennedy's Theorem 1: setup

- Cramér–Rao gives the familiar variance story; Kennedy states a stronger asymptotic minimax version.
- Kennedy starts from a smooth parametric model:

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

- The target is a smooth function of θ :

$$\psi(\theta).$$

- The theorem asks: how small can the local worst-case MSE be for estimating $\psi(\theta)$?

Kennedy's Theorem 1: assumptions

- The score and Fisher information are

$$s_{\theta}(Z) = \frac{\partial}{\partial \theta} \log p_{\theta}(Z), \quad I_{\theta} = \text{Var}_{\theta}[s_{\theta}(Z)].$$

- Assume P_{θ} is differentiable in quadratic mean at θ .
- Assume I_{θ} is nonsingular and $\psi(\theta)$ is differentiable.

Kennedy's Theorem 1: statement

- For any estimator $\hat{\psi}$, Kennedy (2022) states:

$$\inf_{\delta > 0} \liminf_{n \rightarrow \infty} \sup_{\|\theta' - \theta\| < \delta} n \mathbb{E}_{\theta'} \left[\{\hat{\psi} - \psi(\theta')\}^2 \right] \geq \psi'(\theta) I_{\theta}^{-1} \psi'(\theta)^T.$$

- The right-hand side is the parametric efficiency benchmark:

$$\psi'(\theta) I_{\theta}^{-1} \psi'(\theta)^T.$$

- If $\psi(\theta) = \theta$, this reduces to

$$I_{\theta}^{-1}.$$

- So the theorem gives the asymptotic local worst-case MSE bound for smooth functions of θ .

Stronger than Cramér–Rao

- The classical Cramér–Rao bound is often stated for unbiased estimators.
- Kennedy's Theorem 1 is a local asymptotic minimax statement.
- It allows biased estimators, adaptive estimators, unusual estimators, and arbitrary estimator sequences.
- The conclusion is stronger: no estimator can uniformly beat the benchmark in shrinking neighborhoods around θ .
- The bound is about local worst-case asymptotic mean squared error, not just finite-sample variance among unbiased estimators.

From information to influence functions

- Kennedy wants to move from finite-dimensional parametric models to infinite-dimensional nonparametric models.
- In the parametric case, Theorem 1 gives the benchmark

$$\psi'(\theta)I_{\theta}^{-1}\psi'(\theta)^T.$$

- In the semiparametric case, the analogous benchmark is

$$\text{Var} [\varphi(Z; P_0)],$$

where φ is the efficient influence function.

- Parametric submodels provide the bridge: one-dimensional paths through the large model generate Cramér–Rao-type bounds.

Now make it concrete

- In causal ML, the target is often a functional

$$\psi = \Psi(P),$$

not a finite-dimensional parameter θ .

- The unknown distribution P contains outcome regressions, treatment probabilities, covariate distributions, and more.
- Kennedy's Example 2 shows the key objects in a familiar case:

$$\psi(P) = \mathbb{E}_P[\mathbb{E}_P[Y \mid X, A = 1]].$$

- We will use it to see the influence curve and the second-order remainder.

Example 2: expected regression

- Let $Z = (X, A, Y)$, where X are covariates, $A \in \{0, 1\}$ is treatment or missingness, and Y is the outcome.

- Define

$$\mu(x) = \mathbb{E}_P[Y \mid X = x, A = 1], \quad \pi(x) = P[A = 1 \mid X = x].$$

- The target functional is

$$\psi(P) = \mathbb{E}_P[\mathbb{E}_P[Y \mid X, A = 1]] = \mathbb{E}_P[\mu(X)].$$

- Under consistency, positivity, and no unmeasured confounding, this is the statistical functional corresponding to $\mathbb{E}[Y^1]$.
- If $A = 1$ means Y is observed, this is also the mean outcome under missing at random.

Distributional Taylor expansion

- For a smooth functional $\psi(P)$, Kennedy writes

$$\psi(\bar{P}) - \psi(P) = \int \varphi(z; \bar{P}) d(\bar{P} - P)(z) + R_2(\bar{P}, P).$$

- This is the analog of

$$f(\bar{x}) - f(x) = f'(x)(\bar{x} - x) + \text{remainder}.$$

- The analogy is

x	$\leftrightarrow$	P
$\bar{x} - x$	$\leftrightarrow$	$\bar{P} - P$
$f'(x)$	$\leftrightarrow$	$\varphi(z; \bar{P})$
higher-order error	$\leftrightarrow$	$R_2(\bar{P}, P).$

What the expansion means

- The first-order term is

$$\int \varphi(z; \bar{P}) d(\bar{P} - P)(z).$$

- Interpretation:

change in target $\approx$ change in distribution
weighted by influence curve.

- The influence curve $\varphi(z; P)$ tells us how sensitive the target is to putting a little more probability mass near value z .
- The remainder $R_2(\bar{P}, P)$ is second-order: it should involve products or squares of differences between $\bar{P}$ and P .

The expansion for Example 2

- For

$$\psi(P) = \mathbb{E}_P[\mathbb{E}_P[Y \mid X, A = 1]],$$

the influence curve is

$$\varphi(Z; P) = \frac{A}{\pi(X)} \{Y - \mu(X)\} + \mu(X) - \psi(P).$$

- Kennedy states that this satisfies the von Mises expansion with remainder

$$R_2(\bar{P}, P) = \int \left\{ \frac{1}{\bar{\pi}(x)} - \frac{1}{\pi(x)} \right\} \{\bar{\mu}(x) - \mu(x)\} \pi(x) dP(x).$$

- Here $\bar{\mu}(x) = \mathbb{E}_{\bar{P}}[Y \mid X = x, A = 1]$ and $\bar{\pi}(x) = \bar{P}[A = 1 \mid X = x]$.

Two sources of variation

- Since $\psi(P) = \mathbb{E}_P[\mu(X)]$, changing P changes the target in two ways.
- The regression function $\mu(x) = \mathbb{E}_P[Y \mid X = x, A = 1]$ can change.
- The marginal distribution of X can change.

$$\varphi(Z; P) = \underbrace{\frac{A}{\pi(X)} \{Y - \mu(X)\}}_{\text{variation in outcome regression}} + \underbrace{\mu(X) - \psi(P)}_{\text{variation in covariate distribution}} .$$

Derivation: discrete case

- Pretend for the derivation that X is discrete.
- Then

$$\psi(P) = \mathbb{E}_P[\mu(X)] = \sum_x \mu(x)p(x), \quad p(x) = P[X = x].$$

- Use the product rule for influence functions:

$$IF\{\mu(x)p(x)\} = IF\{\mu(x)\}p(x) + \mu(x)IF\{p(x)\}.$$

- Therefore

$$IF\{\psi(P)\} = \sum_x [IF\{\mu(x)\}p(x) + \mu(x)IF\{p(x)\}].$$

Building block 1

- For a point probability $p(x) = P[X = x]$,

$$IF\{p(x)\} = \mathbb{1}(X = x) - p(x).$$

- Therefore the contribution from the marginal distribution of X is

$$\sum_x \mu(x) \{ \mathbb{1}(X = x) - p(x) \}.$$

- Simplify:

$$\sum_x \mu(x) \mathbb{1}(X = x) - \sum_x \mu(x) p(x) = \mu(X) - \psi(P).$$

- So this part gives

$$\boxed{\mu(X) - \psi(P)}.$$

Building block 2

- For $\mu(x) = \mathbb{E}_P[Y \mid X = x, A = 1]$,

$$IF\{\mu(x)\} = \frac{\mathbb{1}(X = x, A = 1)}{P[X = x, A = 1]} \{Y - \mu(x)\}.$$

- Since $P[X = x, A = 1] = p(x)\pi(x)$,

$$IF\{\mu(x)\}p(x) = \frac{\mathbb{1}(X = x, A = 1)}{p(x)\pi(x)} \{Y - \mu(x)\}p(x).$$

- Cancel $p(x)$:

$$IF\{\mu(x)\}p(x) = \frac{\mathbb{1}(X = x, A = 1)}{\pi(x)} \{Y - \mu(x)\}.$$

Summing over x

- The regression contribution is

$$\sum_x \frac{\mathbb{1}(X = x, A = 1)}{\pi(x)} \{Y - \mu(x)\}.$$

- Only the realized value X contributes, so this becomes

$$\frac{A}{\pi(X)} \{Y - \mu(X)\}.$$

- Combining both pieces:

$$\varphi(Z; P) = \frac{A}{\pi(X)} \{Y - \mu(X)\} + \mu(X) - \psi(P).$$

Mean-zero check: residual term

- First term:

$$\mathbb{E}_P \left[\frac{A}{\pi(X)} \{Y - \mu(X)\} \right].$$

- Condition on X :

$$\mathbb{E}_P \left[\frac{A}{\pi(X)} \{Y - \mu(X)\} \mid X \right] = \frac{1}{\pi(X)} \mathbb{E}_P[A\{Y - \mu(X)\} \mid X].$$

- But

$$\mathbb{E}_P[A\{Y - \mu(X)\} \mid X] = \pi(X) \mathbb{E}_P[Y - \mu(X) \mid X, A = 1] = 0.$$

Mean-zero check: covariate term

- The second term is centered by construction:

$$\mathbb{E}_P[\mu(X) - \psi(P)] = \mathbb{E}_P[\mu(X)] - \psi(P) = 0.$$

- Therefore

$$\mathbb{E}_P[\varphi(Z; P)] = 0.$$

- This is the influence-curve analog of the score being mean zero.

Where the remainder comes from

- The first-order expansion removes the linear error.
- What remains is

$$R_2(\bar{P}, P) = \int \left\{ \frac{1}{\bar{\pi}(x)} - \frac{1}{\pi(x)} \right\} \{ \bar{\mu}(x) - \mu(x) \} \pi(x) dP(x).$$

- This has the form

$$R_2(\bar{P}, P) \approx (\bar{\pi} - \pi)(\bar{\mu} - \mu).$$

- So the remainder is a product of two nuisance errors:

propensity score error $\times$ outcome regression error.

Why this remainder matters

- The remainder is not first-order in either nuisance function.
- It is second-order:

$$R_2(\bar{P}, P) \approx (\bar{\pi} - \pi)(\bar{\mu} - \mu).$$

- This is the source of double-robust and debiased behavior.
- If

$$\|\bar{\pi} - \pi\| = o_P(n^{-1/4}), \quad \|\bar{\mu} - \mu\| = o_P(n^{-1/4}),$$

then

$$|R_2(\bar{P}, P)| = o_P(n^{-1/2}).$$

- Then the remainder is negligible relative to the leading root- n sampling error.

The big takeaway

- For $\psi(P) = \mathbb{E}_P[\mathbb{E}_P[Y \mid X, A = 1]]$,

$$\varphi(Z; P) = \frac{A}{\pi(X)} \{Y - \mu(X)\} + \mu(X) - \psi(P).$$

- The term

$$\frac{A}{\pi(X)} \{Y - \mu(X)\}$$

corrects the outcome regression using observed or treated residuals.

- The term

$$\mu(X) - \psi(P)$$

accounts for sampling variation in the covariate distribution.

- The key remainder is

$$R_2(\bar{P}, P) \approx (\bar{\pi} - \pi)(\bar{\mu} - \mu).$$

References I

- Kennedy, E. H. (2022). Semiparametric doubly robust targeted double machine learning: a review. arXiv preprint arXiv:2203.06469.
- Lehmann, E. L. (1999). *Elements of large-sample theory*. Springer.